

Source Name: World Bank

URL:https://data360.worldbank.org/en/indicator/WB_WDI_EG_USE_ELEC_KH_PC?recentYear=false&view=datatable&minYear=1990&maxYear=2024

Data Description:

The dataset "**Electric power consumption (kWh per capita)**" measures the average amount of electrical energy consumed per person in a given country or region over a calendar year, expressed in kilowatt-hours (kWh).

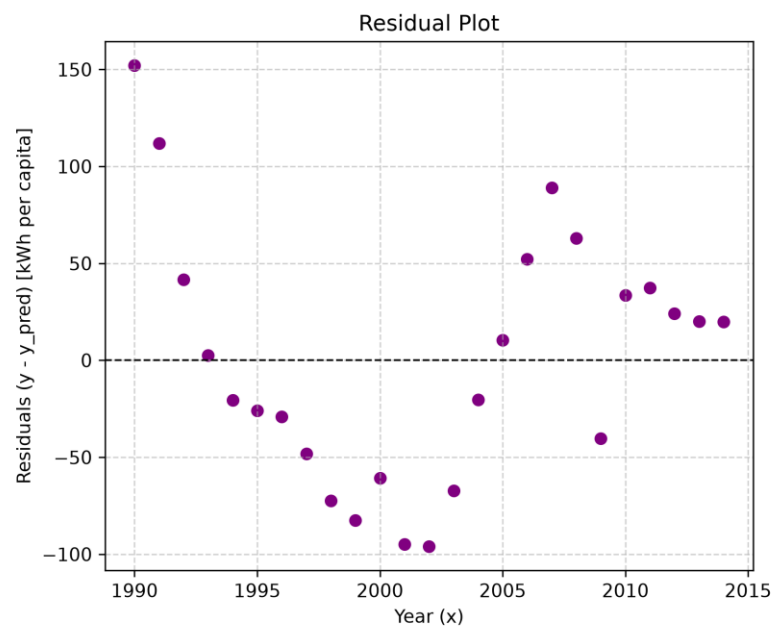
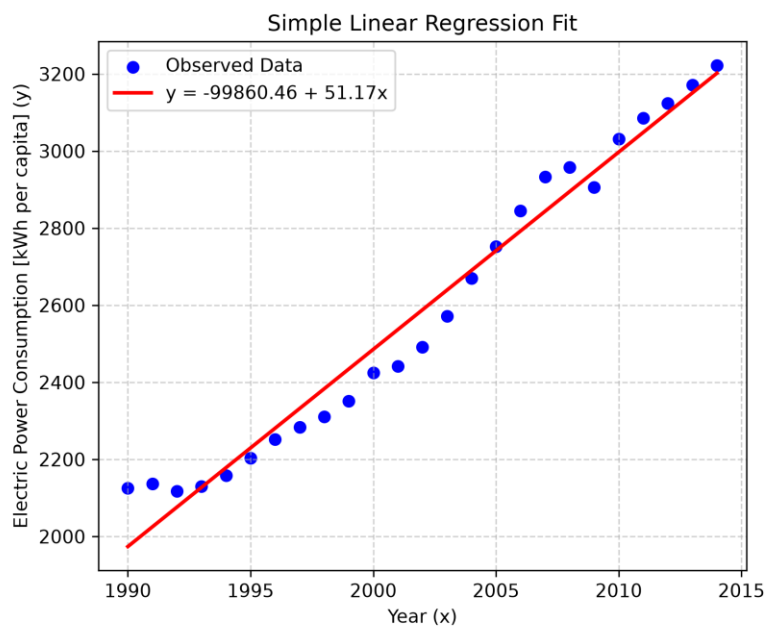
- **What it covers:** It includes electricity generated by power plants and combined heat and power (CHP) plants, accounting for production minus transmission, distribution, and transformation losses, as well as own use by heat and power plants.
- **What it measures:** It serves as a standard metric for assessing energy usage intensity, economic activity, industrialization levels, and general living standards across different economies and time periods.

Independent Variable(x): Time / Year

Dependent Variable(y): Electric Power Consumption per Capita

Year (Year)	Electric Power Consumption (kWh per capita)
1990	2,126
1991	2,137
1992	2,118
1993	2,130
1994	2,158
1995	2,204
1996	2,252
1997	2,284
1998	2,311
1999	2,352
2000	2,425
2001	2,442
2002	2,492
2003	2,572
2004	2,670
2005	2,752
2006	2,845
2007	2,933

2008	2,958
2009	2,906
2010	3,031
2011	3,086
2012	3,124
2013	3,171
2014	3,222



Results Report:

- **Regression Equation:** $\hat{y} = -99,860.46 + 51.1731x$
- **Intercept (a_0):** -99,860.46
- **Slope (a_1):** 51.1731
- **Sum of Squared Errors (S_r / SSE):** 100,782.02 (kWh/capita)²
- **Coefficient of Determination (r^2):** 0.9712
- **Standard Error of Estimate ($s_{y/x}$):** 66.20 kWh per capita

Interpretation of Results:

- **Slope ($a_1 = 51.17 \text{ kWh/capita per year}$):** Represents the estimated average annual increase in global per capita electricity usage between 1990 and 2014. On average, world electricity consumption grew by approximately 51.17 kWh per person each year.
- **Intercept ($a_0 = -99,860.46 \text{ kWh/capita}$):** Represents the theoretical electric power consumption at Year 0 ($x = 0$). Because $x = 0$ lies far outside the observed domain ($1990 \leq x \leq 2014$), the negative intercept has no physical meaning on its own and serves solely as a mathematical anchor for the line.
- **Goodness of Fit (r^2 and $s_{y/x}$):**
 $r^2 = 0.9712$ indicates that 97.12% of the total variance in per capita electricity consumption is explained by a linear trend over time.
The standard error $s_{y/x} = 66.20 \text{ kWh/capita}$ shows that actual observed values typically deviate from the regression line by about $\pm 66.20 \text{ kWh/capita}$.
- **Residual Range:** Residuals range between -95.00 and +152.00 kWh/capita across the 25 observations.

Pattern & Distribution: A slight systematic pattern is visible in the residuals:

1990–1993: Positive residuals (actual consumption was higher than the model's prediction).

1996–2004: Negative residuals (actual consumption dipped below the linear trend).

2005–2014: Positive residuals returning as consumption growth accelerated.

Prediction:

Predicting global electric power consumption per capita for the year $x = 2020$:

- **Equation:** $\hat{y}(2020) = -99,860.46 + 51.173077 \times 2020$
- **Substitution & Result:** $\hat{y}(2020) = -99,860.46 + 103,369.62 = 3,509.16 \text{ kWh per capita}$

This extrapolation provides a linear baseline estimate (3,509.16 kWh per capita) of global energy demand assuming historical 1990–2014 growth rates continued unchanged.

Comparing this model's prediction against actual recorded data for 2020 allows energy analysts to evaluate macro-scale shifts, such as energy efficiency improvements or structural disruptions like the COVID-19 pandemic.